

Constant acceleration kinematics

$$\vec{a} = \frac{d\vec{v}}{dt} \Rightarrow \int \vec{a} dt = \int d\vec{v} \Rightarrow \vec{a}(t_f - t_0) = \vec{v}_f - \vec{v}_0$$

$$\Rightarrow \boxed{\vec{v}(t) = \vec{v}(0) + \vec{a}t} \quad (1)$$

Integrate with respect to t again:

$$\int_0^t \vec{v}(t') dt' = \int_0^t (\vec{v}(0) + \vec{a}t') dt'$$

$$\Rightarrow \int \frac{d\vec{x}}{dt} dt = \vec{v}(0)t + \frac{1}{2} \vec{a}t^2 \quad (\vec{v}(0), \vec{a} \text{ independent of time})$$

$$\Rightarrow \vec{x}(t) - \vec{x}(0) = \vec{v}(0)t + \frac{1}{2} \vec{a}t^2$$

$$\text{OR } \boxed{\vec{x}(t) = \vec{x}(0) + \vec{v}(0)t + \frac{1}{2} \vec{a}t^2} \quad (2)$$

Now, the difference in position between times zero and t is

$$\vec{x}(t) - \vec{x}(0) = \vec{v}_{\text{avg}} t = \frac{1}{2} (\vec{v}(0) + \vec{v}(t)) t$$

Here $\vec{v}_{\text{avg}}$ is the average velocity over the time interval $[0, t]$.

Then, using (1), and noting these are dot products:

$$\vec{a} \cdot (\vec{x}(t) - \vec{x}(0)) = \frac{\vec{v}(t) - \vec{v}(0)}{t} \cdot \frac{1}{2} (\vec{v}(0) + \vec{v}(t)) t$$

$$\Rightarrow \vec{a} \cdot \Delta \vec{x}(t) = \frac{1}{2} (|\vec{v}(t)|^2 - |\vec{v}(0)|^2 + \vec{v}(t) \cdot \vec{v}(0) - \vec{v}(0) \cdot \vec{v}(t))$$

$$\Rightarrow 2 \vec{a} \cdot \Delta \vec{x}(t) = |\vec{v}(t)|^2 - |\vec{v}(0)|^2$$

$$\text{OR } \boxed{|\vec{v}_f|^2 - |\vec{v}_0|^2 = 2 \vec{a} \cdot \Delta \vec{x}} \quad (3)$$

In 1D this simplifies to $v_f^2 - v_0^2 = 2a \Delta x$